AQA 3.1.4 and 3.1.10 Equilibrium constants K_c and K_p

AQA A-level Chemistry · Revision questions · Sheet 5 of 9

Answer in AQA exam style: use the marks as a guide to how much to write and show all working in calculations. Mark your work with the mark scheme at the back. Questions marked as practice questions in the mark scheme were written in AQA style; all others are from AQA past papers.

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21 marks · about 25 minutes

Key information

- K_p uses partial pressures in round brackets, e.g. $K_p = p(C)^2 \div p(A)\ p(B)^3$; never square brackets
- Find equilibrium amounts with a start / change / equilibrium table using the mole ratio in the equation
- K_c: divide each amount by the volume in dm^3 before substituting (unless the volumes cancel)
- Only temperature changes the value of K_p or K_c; pressure, concentration and catalysts do not
- Exothermic forward reaction: raising temperature lowers K; endothermic: raising temperature increases K

Section A: Short-answer questions

01 Give an expression for K_p for the equilibrium $2NO(g) + O_2(g) \rightleftharpoons 2NO_2(g)$

[1 mark]

02 Which change leads to a higher concentration of SO_3 in this equilibrium mixture?

$2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g) \quad \Delta H = -188\ kJ\ mol^{-1}$

[1 mark]

A higher concentration of O_2
B higher temperature
C lower pressure
D use of a catalyst

03 For the equilibrium $2A(g) \rightleftharpoons B(g)$, which are the units of K_p when pressures are in kPa?

[1 mark]

A kPa
B kPa^{-1}
C kPa^2
D no units

Section B: Longer questions

04 Tetrafluoroethene is made from chlorodifluoromethane in this reversible reaction.

$2CHClF_2(g) \rightleftharpoons C_2F_4(g) + 2HCl(g) \quad \Delta H = +128\ kJ\ mol^{-1}$

A 2.00 mol sample of $CHClF_2$ is placed in a container of volume 23.2 dm^3 and heated. When equilibrium is reached, the mixture contains 0.270 mol of $CHClF_2$.

04.1 Calculate the amount, in moles, of C_2F_4 and of HCl in the equilibrium mixture.

[2 marks]

04.2 Give an expression for K_c for this equilibrium.

[1 mark]

04.3 Calculate a value for K_c. Give its units.

[3 marks]

04.4 State and explain the effect of using a higher temperature on the equilibrium yield of tetrafluoroethene.

[3 marks]

05 This question is about chemical equilibria.

05.1 A mixture of hydrogen and iodine is allowed to reach equilibrium in a sealed container.

$H_2(g) + I_2(g) \rightleftharpoons 2HI(g) \quad \Delta H = +52 \text{ kJ mol}^{-1}$

State and explain the effect of increasing the temperature on the equilibrium yield of hydrogen iodide.

[3 marks]

05.2 Carbon dioxide can react with hydrogen to form methanol.

$CO_2(g) + 3H_2(g) \rightleftharpoons CH_3OH(g) + H_2O(g)$

2.50 mol of CO_2 are mixed with 6.00 mol of H_2 in a sealed container with a volume of 4.60 dm^3. The mixture is left to reach equilibrium at a fixed temperature. At equilibrium, 1.30 mol of CO_2 remain.

Give an expression for K_c for this reaction. Calculate a value for the equilibrium constant at this fixed temperature. Give the units of the equilibrium constant.

[6 marks]

Mark Scheme

M1, M2… are separate marking points. / separates acceptable alternatives. Words in brackets are not needed for the mark. ECF = error carried forward. Questions and mark schemes © AQA, reproduced for internal school use.

Question	Marking guidance	Comments	Mark
01	$K_p = p(NO_2)^2 \div (p(NO)^2 \times p(O_2))$	*Source: Practice question (written for this set)* Penalise square brackets	1
02	**A**	*Source: 7405/3 Jun 2019 Q09*	1
03	**B**	*Source: Practice question (written for this set)*	1
04.1	C_2F_4 = 0.865 mol HCl = 1.73 mol	*Source: 7405/2 Jun 2022 Q02.1–02.4* Award 1 mark if HCl = 2 × C_2F_4	2
04.2	$K_c = [C_2F_4][HCl]^2 \div [CHClF_2]^2$	Penalise round brackets	1
04.3	M1 $K_c = (0.865/23.2)(1.73/23.2)^2 \div (0.27/23.2)^2$ (dividing by volume) M2 K_c = 1.5(3) M3 mol dm^{-3}	Allow 1.53–1.54; at least 2 sf If no use of volume, allow M2 for 35.5	3
04.4	M1 yield would increase M2 equilibrium opposes temperature increase / shifts to reduce temperature M3 moves in the endothermic direction	Ignore favours	3
05.1	M1 yield would increase M2 equilibrium shifts in the endothermic direction (to the right) M3 to oppose the temperature increase / to decrease the temperature	*Source: 7404/1 Jun 2025 Q07.2–07.3* Mark each point independently	3
05.2	M1 $K_c = [CH_3OH][H_2O] \div [CO_2][H_2]^3$ M2 $n(CO_2)$ reacted = 1.20 mol M3 $n(H_2)$ = 2.40 mol, $n(CH_3OH)$ = 1.20 mol, $n(H_2O)$ = 1.20 mol M4 concentrations: $[CH_3OH] = [H_2O]$ = 0.261, $[H_2]$ = 0.522, $[CO_2]$ = 0.283 mol dm^{-3} M5 $K_c = 0.261 \times 0.261 \div (0.283 \times 0.522^3)$ = 1.69 / 1.7(0) M6 mol^{-2} dm^6	Do not accept round brackets in expression	6